

NAME: SAGAR LADLA

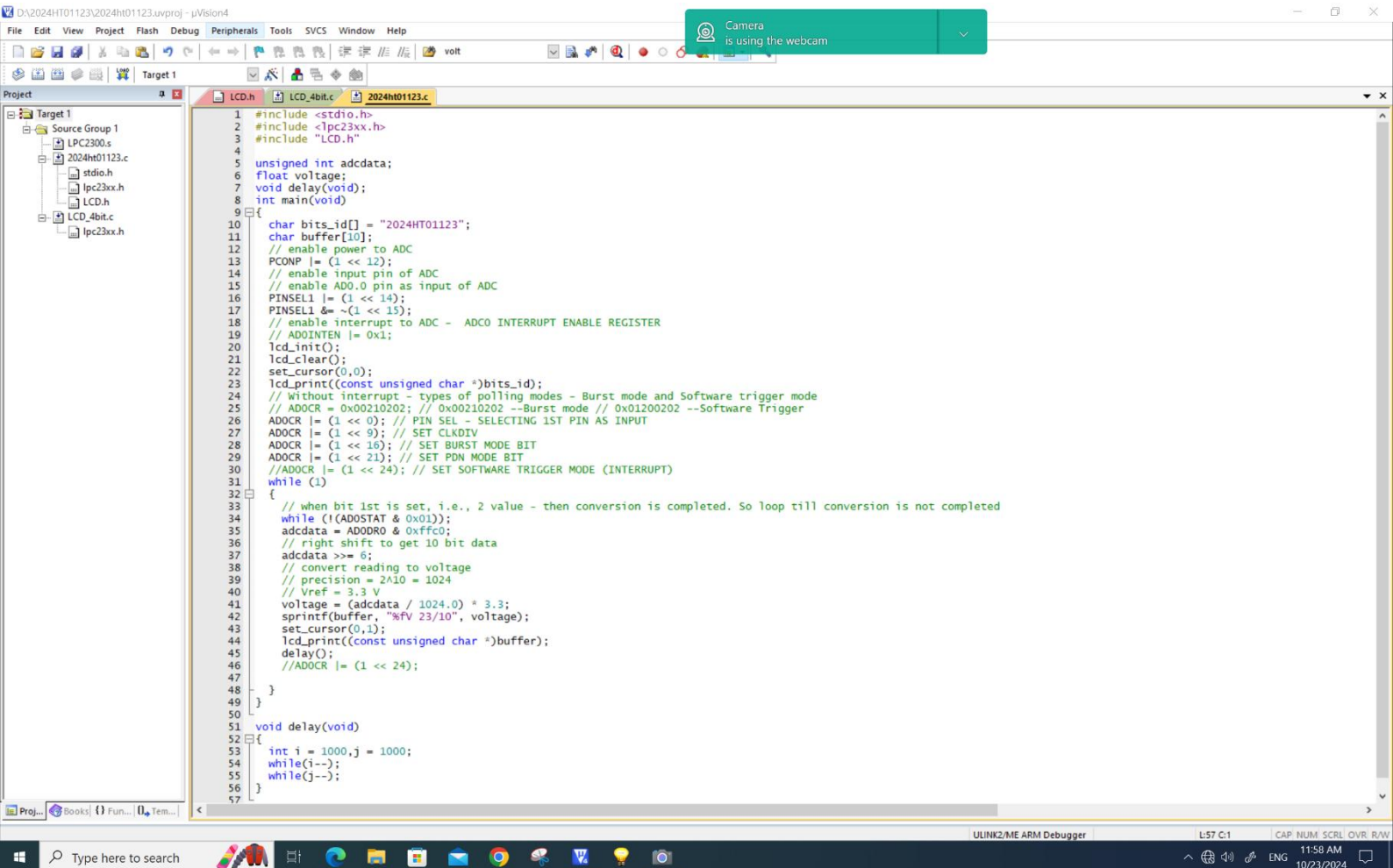
ROLL: 2024HT01123

SUBJECT: EMBEDDED SYSTEM DESIGN

LAB ASSIGNMENT 2

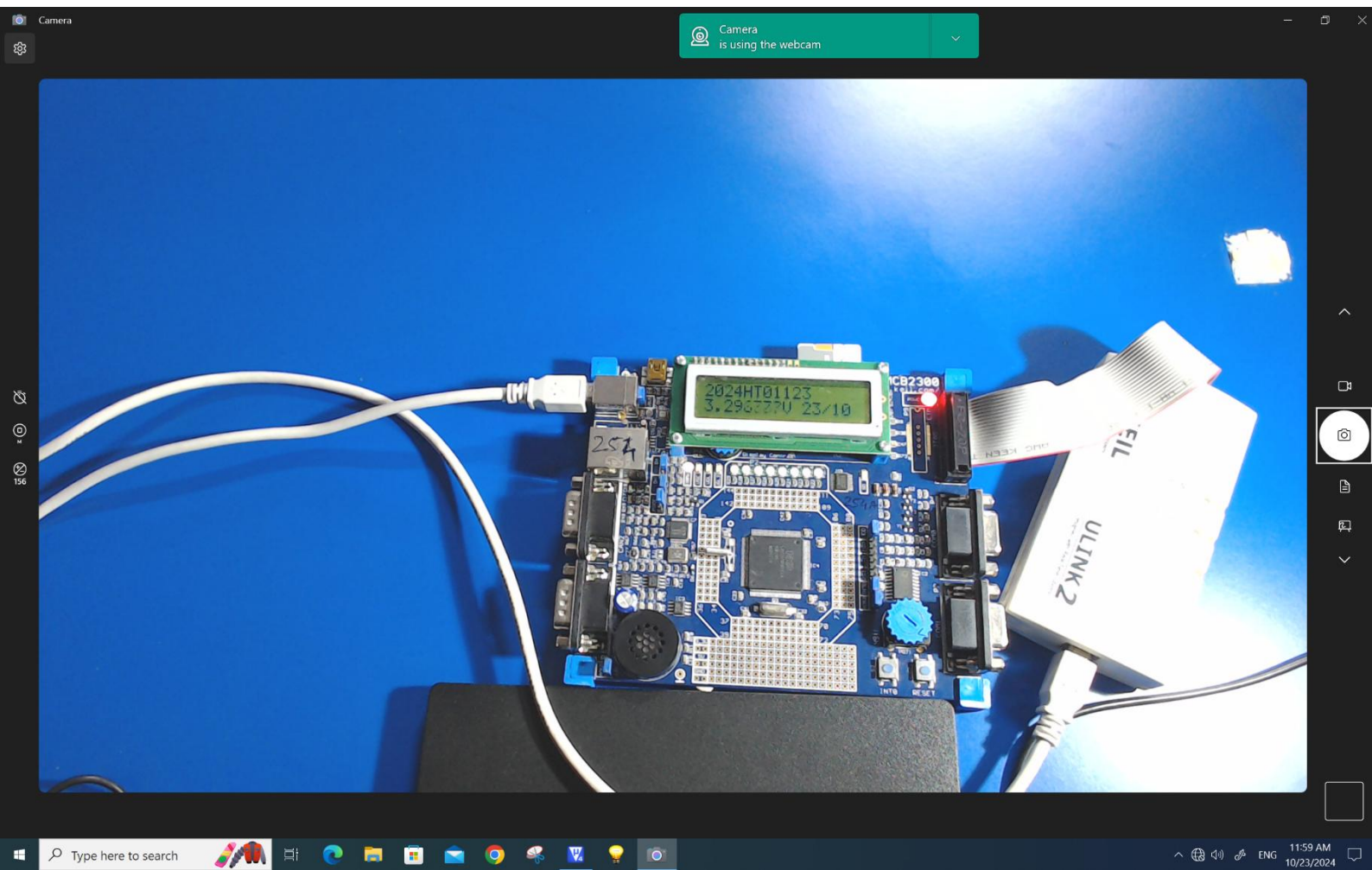
ANS 1

CODE WITH HIGH PRECISION VOLTAGE MEASURE:

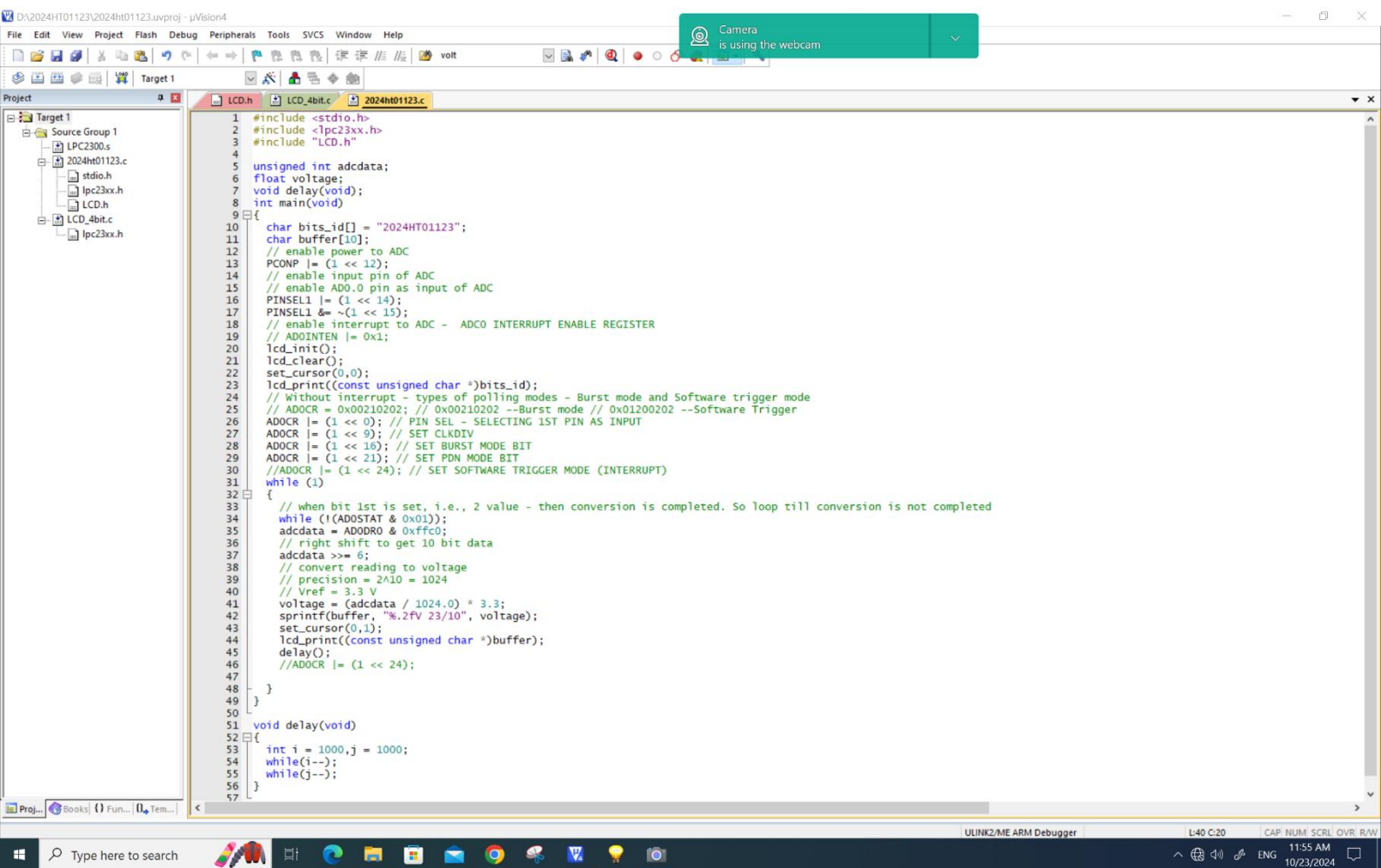


```
1 #include <stdio.h>
2 #include <lpc23xx.h>
3 #include "LCD.h"
4
5 unsigned int adcddata;
6 float voltage;
7 void delay(void);
8 int main(void)
9 {
10     char bits_id[] = "2024HT01123";
11     char buffer[10];
12     // enable power to ADC
13     PCONP |= (1 << 12);
14     // enable input pin of ADC
15     // enable ADO.0 pin as input of ADC
16     PINSEL1 |= (1 << 14);
17     PINSEL1 &= ~(1 << 15);
18     // enable interrupt to ADC - ADC0 INTERRUPT ENABLE REGISTER
19     // AD0INTEN |= 0x1;
20     lcd_init();
21     lcd_clear();
22     set_cursor(0,0);
23     lcd_print((const unsigned char *)bits_id);
24     // Without interrupt - types of polling modes - Burst mode and Software trigger mode
25     // ADCR = 0x00210202; // 0x00210202 --Burst mode // 0x01200202 --Software Trigger
26     ADCR |= (1 << 0); // PIN SEL - SELECTING 1ST PIN AS INPUT
27     ADCR |= (1 << 9); // SET CLKDIV
28     ADCR |= (1 << 16); // SET BURST MODE BIT
29     ADCR |= (1 << 21); // SET PDN MODE BIT
30     //ADOCR |= (1 << 24); // SET SOFTWARE TRIGGER MODE (INTERRUPT)
31     while (1)
32     {
33         // when bit 1st is set, i.e., 2 value - then conversion is completed. So loop till conversion is not completed
34         while (!(AD0STAT & 0x01));
35         adcddata = AD0DR0 & 0xffc0;
36         // right shift to get 10 bit data
37         adcddata >>= 6;
38         // convert reading to voltage
39         // precision = 2A10 = 1024
40         // Vref = 3.3 V
41         voltage = (adcddata / 1024.0) * 3.3;
42         sprintf(buffer, "%FV 23/10", voltage);
43         set_cursor(0,1);
44         lcd_print((const unsigned char *)buffer);
45         delay();
46         //ADOCR |= (1 << 24);
47     }
48 }
49
50
51 void delay(void)
52 {
53     int i = 1000, j = 1000;
54     while(i--);
55     while(j--);
56 }
57
```

OUTPUT WITH HIGH PRECISION VOLTAGE MEASURE:

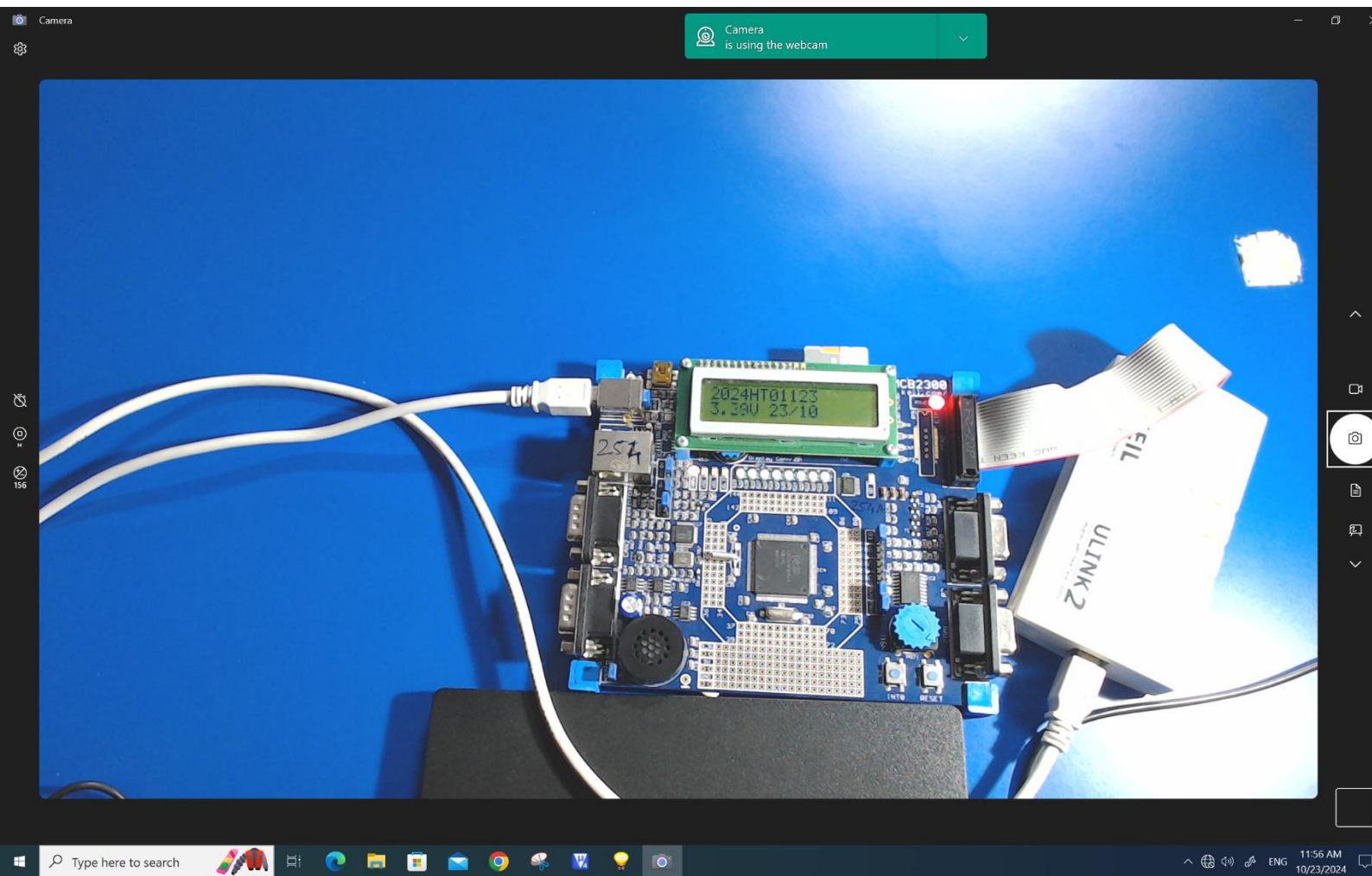


CODE WITH FIXED POINT VALUE:



```
1 #include <stdio.h>
2 #include <lpc23xx.h>
3 #include "LCD.h"
4
5 unsigned int adcddata;
6 float voltage;
7 void delay(void);
8 int main(void)
9 {
10     char bits_id[] = "2024HT01123";
11     char buffer[10];
12     // enable power to ADC
13     PCOMP |= (1 << 12);
14     // enable input pin of ADC
15     // enable ADO.0 pin as input of ADC
16     PINSEL1 |= (1 << 14);
17     PINSEL1 &= ~(1 << 15);
18     // enable interrupt to ADC - ADC0 INTERRUPT ENABLE REGISTER
19     // AD0INTEN |= 0x1;
20     lcd_init();
21     lcd_clear();
22     set_cursor(0,0);
23     lcd_print((const unsigned char *)bits_id);
24     // Without interrupt - types of polling modes - Burst mode and Software trigger mode
25     // ADCR = 0x00210202; // 0x00210202 --Burst mode // 0x01200202 --Software Trigger
26     ADCR |= (1 << 0); // PIN SEL - SELECTING 1ST PIN AS INPUT
27     ADCR |= (1 << 9); // SET CLKDIV
28     ADCR |= (1 << 16); // SET BURST MODE BIT
29     ADCR |= (1 << 21); // SET PDN MODE BIT
30     //ADOCR |= (1 << 24); // SET SOFTWARE TRIGGER MODE (INTERRUPT)
31     while (1)
32     {
33         // when bit 1st is set, i.e., 2 value - then conversion is completed. So loop till conversion is not completed
34         while (!(AD0STAT & 0x01));
35         adcddata = AD0DR0 & 0xffc0;
36         // right shift to get 10 bit data
37         adcddata >>= 6;
38         // convert reading to voltage
39         // precision = 2^10 = 1024
40         // Vref = 3.3 V
41         voltage = (adcddata / 1024.0) * 3.3;
42         sprintf(buffer, "%.2FV 23/10", voltage);
43         set_cursor(0,1);
44         lcd_print((const unsigned char *)buffer);
45         delay();
46         //ADOCR |= (1 << 24);
47     }
48 }
49
50 void delay(void)
51 {
52     int i = 1000, j = 1000;
53     while(i--);
54     while(j--);
55 }
56
57
```

OUTPUT WITH FIXED POINT VALUE:



ANS 2:

- a) Smallest change in input voltage detected by LPC2378 ADC (Resolution),
 - i. $R = V_{\text{ref}} / 2^n$
 - ii. $R = 3.3 \text{ V} / 2^{10}$
 - iii. $R = 3.3 \text{ V} / 1024$
 - iv. **$R = 0.0032226563 \text{ V}$ or $R = 3.2226563 \text{ mV}$**
- b) Operating f_{max} of LPC2378 ADC = **4.5 Mhz**
- c) Steps to program timer for 2 second delay:
 - i. Reset timer0 initially to deactivate counting.
 - i. $\text{TOTCR} = (1 \ll 1);$
 - ii. Load calculated values in the Prescaler register TOPR and Match register TOMRO
 - i. $\text{TOMRO} = 599999$
 - ii. $\text{TOPR} = 10$
 - iii. Initialize TOPC and TOTC registers.
 - i. $\text{TOPC} = 0;$
 - ii. $\text{TOTC} = 0;$
 - iv. Select operations using match registers when match is encountered.
 - i. $\text{TOMCR} = (1 \ll 0) | (1 \ll 2);$
 - v. Start the Timer by enabling it through TOTCR register.
 - i. $\text{TOTCR} = (1 \ll 0);$
 - vi. Wait till the interrupt and then clear the flag by writing TOIR register.
 - i. $\text{while}(!(\text{TOIR} \& (1 \ll 0)));$
 - ii. $\text{TOIR} = (1 \ll 0);$

CCLK = 48 MHz (given)

So, by using default $\frac{1}{4}$ divider, we will the PCLK = $48 \text{ MHz} / 4 = 12 \text{ MHz}$

Therefore, count:

$$= (\text{PCLK} / \text{Desired frequency}) - 1$$

$$= (12,000,000 / 20) - 1$$

$$= 6,00,000 - 1$$

$$\text{Count} = 5,99,999$$

$\text{TOMRO} = 5,99,999$ is count which gives 0.2 seconds delay. To make the delay for 2 seconds we have to multiply it with 10, which we set in TOPR (prescaler) register.

Full code of LCD-ADC CODE:

```
#include <stdio.h>

#include <|pc23xx.h>

#include "LCD.h"

unsigned int adcddata;

float voltage;

void delay(void);

int main(void)
{
    char bits_id[] = "2024HT01123";
    char buffer[10];

    // enable power to ADC
    PCONP |= (1 << 12);

    // enable input pin of ADC
    // enable AD0.0 pin as input of ADC
    PINSEL1 |= (1 << 14);
    PINSEL1 &= ~(1 << 15);

    // enable interrupt to ADC -    ADC0 INTERRUPT ENABLE REGISTER
    // AD0INTEN |= 0x1;

    lcd_init();
    lcd_clear();
    set_cursor(0,0);
    lcd_print((const unsigned char *)bits_id);

    // Without interrupt - types of polling modes - Burst mode and Software trigger mode
    // AD0CR = 0x00210202; // 0x00210202 --Burst mode // 0x01200202 --Software Trigger
    AD0CR |= (1 << 0); // PIN SEL - SELECTING 1ST PIN AS INPUT
    AD0CR |= (1 << 9); // SET CLKDIV
    AD0CR |= (1 << 16); // SET BURST MODE BIT
    AD0CR |= (1 << 21); // SET PDN MODE BIT
    //AD0CR |= (1 << 24); // SET SOFTWARE TRIGGER MODE (INTERRUPT)
```

```

while (1)
{
    // when bit 1st is set, i.e., 2 value - then conversion is completed. So loop till
conversion is not completed

    while (!(ADOSTAT & 0x01));

    adcddata = AD0DR0 & 0xffc0;

    // right shift to get 10 bit data
    adcddata >>= 6;

    // convert reading to voltage
    // precision =  $2^{10} = 1024$ 
    // Vref = 3.3 V
    voltage = (adcddata / 1024.0) * 3.3;
    sprintf(buffer, "%.2fV 23/10", voltage);
    set_cursor(0,1);
    lcd_print((const unsigned char *)buffer);
    delay();

    //AD0CR |= (1 << 24);

}
}

```

```

void delay(void)
{
    int i = 1000,j = 1000;
    while(i--);
    while(j--);
}

```